

1.1. A partir de

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu, \quad (1.1)$$

muestre que es posible recuperar la ley de Gauss y la ley de Ampère-Maxwell.

⇒ Calculemos qué ocurre cuando $v=0$

$$\partial_\mu F^{\mu 0} = \mu_0 J^0 \quad (i)$$

expandiendo la sumatoria de forma explícita:

$$\partial_\mu F^{\mu 0} = \partial_0 F^{00} + \partial_1 F^{10} + \partial_2 F^{20} + \partial_3 F^{30}$$

al ser el tensor de Faraday un tensor anti-simétrico $\Rightarrow F^{00}=0$

$$\begin{aligned} \partial_\mu F^{\mu 0} &= \partial_1 F^{10} + \partial_2 F^{20} + \partial_3 F^{30} \\ &= \partial_i F^{i0} \end{aligned}$$

Calculemos F^{i0}

$$F_{i0} = \partial_0 A_i - \partial_i A_0 \quad / \quad \text{como } A_\mu = (\varphi, -\vec{A})$$

$$= -\partial_0 (\vec{A})_i - \partial_i \varphi$$

$$= \left(-\frac{dA}{dt} - \vec{\nabla} \varphi \right)_i = \frac{E_i}{c}$$

$$F_{i0} = \frac{E_i}{c} \quad / \quad \partial_i$$

$$\partial_i F_{i0} = \partial_i E_i = \frac{1}{c} \vec{\nabla} \cdot \vec{E} \quad (ii)$$

Para la 4-corriente, el término $J^\mu = (c\rho, \vec{j})$ ↗ densidad de carga
↘ densidad de corriente

Entonces, evaluando cuando $\mu=0 \Rightarrow \mu_0 J^0 = \mu_0 c\rho \quad (iii)$

Reemplazando las ec. (ii) y (iii) en (i), obtendremos:

$$\partial_\mu F^{\mu 0} = \mu_0 J^0$$

$$\frac{1}{c} (\vec{\nabla} \cdot \vec{E}) = \mu_0 c\rho \quad / \quad \frac{1}{c^2} = \mu_0 \epsilon_0$$

$$\boxed{\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}} \quad \text{recuperamos la ley de Gauss.}$$

⇒ Calculemos ahora cuando $v=i$. Obtendremos que:

$$\partial_\mu F^{\mu i} = \partial_0 F^{0i} + \partial_1 F^{1i} + \partial_2 F^{2i} + \partial_3 F^{3i} \quad i = 1, 2, 3$$

• Para la derivada temporal:

$$\partial_0 F^{0i} = \frac{1}{c} \frac{\partial}{\partial t} \left(-\frac{E_i}{c} \right) = -\frac{1}{c^2} \frac{\partial E_i}{\partial t} \quad (iv)$$

• Para las derivadas espaciales, podemos usar la ec. (0.4), la cual nos indica que:

$$F_{ij} = -\epsilon_{ijk} B_k$$

en la notación que tenemos:

$$F^{ji} = -\epsilon^{jik} B_k / \partial_j$$

$$\partial_j F^{ji} = -\epsilon^{jik} \partial_j B_k$$

considerando la asimetría del símbolo de Levi-Civita: $\epsilon^{jik} = -\epsilon^{ijk}$

$$\Rightarrow \partial_j F^{ji} = \epsilon^{ijk} \partial_j B_k$$

y la componente i -ésima del rotacional está definida como: $(\nabla \times B)^i = \epsilon^{ijk} \partial_j B_k$

$$\partial_j F^{ji} = (\nabla \times B)^i \quad (v)$$

Para la corriente: $\mu_0 J_i$ (vi)

Uniendo (iv), (v) y (vi):

$$\partial_\mu F^{\mu i} = \mu_0 J^i$$

$$-\frac{1}{c^2} \frac{\partial E_i}{\partial t} + (\nabla \times B)^i = \mu_0 J^i$$

$$(\nabla \times B)^i = \mu_0 J^i + \frac{1}{c^2} \frac{\partial E_i}{\partial t}$$

$$\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \frac{\partial E}{\partial t}$$

Recuperamos la ec. de Ámper - Maxwell

1.2. Calcule las componentes de

$$\partial_\mu \tilde{F}^{\mu\nu} = 0, \quad (1.2)$$

y muestre que se recuperan las ecuaciones homogéneas de Maxwell.

Aparece $\tilde{F}^{\mu\nu}$, el cual corresponde al tensor dual de Faraday (dual de Hodge del tensor EM). Intercambia los campos $\vec{E}$ y $\vec{B}$.

Similar al ejercicio anterior, evaluamos por casos.

$\Rightarrow$ Evaluemos cuando $\nu=0$

$$\partial_\mu \tilde{F}^{\mu 0} = \partial_0 \tilde{F}^{00} + \partial_1 \tilde{F}^{10} + \partial_2 \tilde{F}^{20} + \partial_3 \tilde{F}^{30}$$

$$= \partial_i \tilde{F}^{i0} \quad (vii)$$

usando las ec. (0.3) y (0.4):

$$\begin{aligned}\tilde{F}^{10} &= \frac{1}{2} \epsilon^{10jk} F_{jk} \quad / \quad F_{jk} = -\epsilon_{jkl} B_l \\ &= -\frac{1}{2} \epsilon^{10jk} (-\epsilon_{jkl} B_l) \quad / \quad \epsilon^{10jk} = -\epsilon^{ijk} \\ &= \frac{1}{2} \epsilon^{ijk} \epsilon_{jkl} B_l\end{aligned}$$

$$\tilde{F}^{10} = \frac{1}{2} (\epsilon^{ijk} \epsilon_{jkl}) B_l = B_i$$

reemplazamos en la ec. (vii) :

$$\begin{aligned}\partial_\mu \tilde{F}^{\mu 0} &= \partial_i B_i \\ &= \nabla \cdot \mathbf{B} = 0\end{aligned}$$

$$\Rightarrow \boxed{\nabla \cdot \mathbf{B} = 0}$$

$\Rightarrow$ Evaluemos cuando $\nu=1$

$$\partial_0 \tilde{F}^{0i} = \frac{1}{c} \frac{\partial}{\partial t} (-B_i) \quad (\text{viii})$$

$$\partial_\mu \tilde{F}^{\mu i} = \partial_0 \tilde{F}^{0i} + \partial_j \tilde{F}^{ji}$$

veamos qué ocurre cuando $\partial_j \tilde{F}^{ji}$. Calculemos primero el dual :

$$\tilde{F}^{ji} = \frac{1}{2} (\epsilon^{j10k} F_{0k} + \epsilon^{j1k0} F_{k0})$$

considerando que $F_{k0} = -F_{0k}$ y $\epsilon^{ij'k0} = -\epsilon^{ij0k}$

$$\tilde{F}^{ji} = \frac{1}{2} (\epsilon^{j10k} F_{0k} + \epsilon^{j10k} F_{0k})$$

$$\tilde{F}^{ji} = \epsilon^{j10k} F_{0k} = \epsilon^{jik} F_{0k} \quad \text{del inciso 1.1 : } F_{0k} = \frac{E_k}{c}$$

$$\tilde{F}^{ji} = \frac{1}{c} \epsilon^{jik} E_k \quad / \quad \partial_j$$

$$\partial_j \tilde{F}^{ji} = \frac{1}{c} \epsilon^{jik} \partial_j E_k \quad / \quad \epsilon^{jik} = -\epsilon^{ijk}$$

$$= -\frac{1}{c} \epsilon^{ijk} \partial_j E_k$$

$$\partial_j \tilde{F}^{ji} = -\frac{1}{c} (\vec{\nabla} \times \vec{E})_i \quad (\text{ix})$$

Sumando las ec. (viii) y (ix) :

$$-\frac{1}{c} \frac{\partial B_i}{\partial t} - \frac{1}{c} (\vec{\nabla} \times \vec{E})_i = 0$$

$$\boxed{\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}}$$

2.1. Considere la densidad lagrangiana

$$\mathcal{L} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - J_\mu A^\mu. \quad (2.1)$$

Varíe la acción respecto de A_μ y obtenga las ecuaciones de Maxwell

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu. \quad (2.2)$$

Discuta en qué paso aparece la conservación de la carga,

$$\partial_\mu J^\mu = 0. \quad (2.3)$$

La acción de esta lagrangiana será:

$$S = \int d^4x \left[-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - J_\mu A^\mu \right]$$

recordando que: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ / δ

$$\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu$$

$$\Rightarrow \delta S = \int d^4x \left[-\frac{1}{4\mu_0} \delta(F_{\mu\nu} F^{\mu\nu}) - \delta(J_\mu A^\mu) \right]$$

$$= \int d^4x \left[-\frac{1}{4\mu_0} \left\{ (\delta F_{\mu\nu}) F^{\mu\nu} + F_{\mu\nu} (\delta F^{\mu\nu}) \right\} - J^\nu \delta A_\nu \right]$$

$$\left. \begin{aligned} F^{\mu\nu} &= \eta^{\mu\alpha} \eta^{\nu\beta} F_{\alpha\beta} / \delta \Rightarrow \delta F^{\mu\nu} = \eta^{\mu\alpha} \eta^{\nu\beta} \delta F_{\alpha\beta} \\ F_{\mu\nu} &= \eta_{\mu\alpha} \eta_{\nu\beta} F^{\alpha\beta} / \delta \Rightarrow \delta F_{\mu\nu} = \eta_{\mu\alpha} \eta_{\nu\beta} \delta F^{\alpha\beta} \end{aligned} \right\} F_{\mu\nu} \delta F^{\mu\nu} = F^{\mu\nu} \delta F_{\mu\nu}$$

$$= \int d^4x \left\{ \left[-\frac{1}{2\mu_0} F^{\mu\nu} \delta F_{\mu\nu} \right] - J^\nu \delta A_\nu \right\}$$

$$= \frac{1}{2\mu_0} \int d^4x \left\{ (F^{\mu\nu} \delta F_{\mu\nu}) - J^\nu \delta A_\nu \right\}$$

$$= \frac{1}{2\mu_0} \int d^4x \left(\underline{F^{\mu\nu} (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu)} - \int d^4x J^\nu \delta A_\nu \right)$$

analicemos esto

$$F^{\mu\nu} \partial_\mu \delta A_\nu - F^{\mu\nu} \partial_\nu \delta A_\mu = F^{\mu\nu} \partial_\mu \delta A_\nu - F^{\nu\mu} \partial_\mu \delta A_\nu \quad / -F^{\nu\mu} = F^{\mu\nu}$$

$$= F^{\mu\nu} \partial_\mu \delta A_\nu + F^{\mu\nu} \partial_\mu \delta A_\nu$$

$$= 2 F^{\mu\nu} \partial_\mu \delta A_\nu$$

$$= \underline{\frac{-1}{\mu_0} \int d^4x F^{\mu\nu} \partial_\mu \delta A_\nu} - \int d^4x J^\nu \delta A_\nu$$

integración por partes.

$$\partial_\mu (F^{\mu\nu} \delta A_\nu) = (\partial_\mu F^{\mu\nu}) \delta A_\nu + F^{\mu\nu} \partial_\mu \delta A_\nu$$

$$\Rightarrow F^{\mu\nu} \partial_\mu \delta A_\nu = \partial_\mu (F^{\mu\nu} \delta A_\nu) - (\partial_\mu F^{\mu\nu}) \delta A_\nu$$

$$- \int d^4x F^{\mu\nu} \partial_\mu \delta A_\nu = - \int d^4x \partial_\mu (F^{\mu\nu} \delta A_\nu) + \int d^4x (\partial_\mu F^{\mu\nu}) \delta A_\nu$$

$$- \int d^4x F^{\mu\nu} \partial_\mu \delta A_\nu = \int d^4x (\partial_\mu F^{\mu\nu}) \delta A_\nu$$

$$\Rightarrow \delta S = \int d^4x \left[\frac{1}{\mu_0} \partial_\mu F^{\mu\nu} - J^\nu \right] \delta A_\nu$$

solución no trivial: $\frac{1}{\mu_0} \partial_\mu F^{\mu\nu} - J^\nu = 0$

$$\Rightarrow \partial_\mu F^{\mu\nu} = \mu_0 J^\nu$$

$$\partial_\nu \partial_\mu F^{\mu\nu} = \mu_0 \partial_\nu J^\nu$$

$$0 = \mu_0 \partial_\nu J^\nu$$

$$\Rightarrow \partial_\nu J^\nu = 0$$

También puede demostrarse con la invarianza de gauge del término $-J_\mu A^\mu$

2.2. Considere la teoría libre definida por

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad (2.4)$$

con

$$F_{\mu\nu} = 2\partial_{[\mu} A_{\nu]} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (2.5)$$

Muestre que la acción es invariante bajo las transformaciones de gauge

$$A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \Lambda(x), \quad (2.6)$$

con $\Lambda(x)$ un campo escalar arbitrario.

Conocemos que $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

$$F'_{\mu\nu} = \partial_\mu (A_\nu + \partial_\nu \Lambda) - \partial_\nu (A_\mu + \partial_\mu \Lambda)$$

$$= F_{\mu\nu} + \partial_\mu \partial_\nu \Lambda - \partial_\nu \partial_\mu \Lambda$$

recordando que las derivadas parciales conmutan $\partial_\mu \partial_\nu = \partial_\nu \partial_\mu$

$$F'_{\mu\nu} = F_{\mu\nu} + \cancel{\partial_\mu \partial_\nu \Lambda} - \cancel{\partial_\nu \partial_\mu \Lambda}$$

$$F'_{\mu\nu} = F_{\mu\nu}$$

entonces $F'_{\mu\nu} F'^{\mu\nu} = F_{\mu\nu} F^{\mu\nu}$. Reemplazando en la acción:

$$\underbrace{S = -\frac{1}{4} \int d^4x F_{\mu\nu} F^{\mu\nu}}_{\text{es invariante bajo transformaciones de gauge}}$$

2.3. Agregue a la densidad lagrangiana de Maxwell un término de masa para el fotón,

$$\mathcal{L} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} + \frac{\mu_0^2}{2} A_\mu A^\mu. \quad (2.7)$$

Derive las ecuaciones de Proca y explique por qué este término rompe la invariancia de gauge.

Recordemos a las ec. de Euler-Lagrange

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) = 0$$

en nuestro caso $\phi_a = A_\nu$

$$\frac{\partial \mathcal{L}}{\partial A_\nu} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) = 0$$

para el primer término:

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = \frac{\partial}{\partial A_\nu} \left(\frac{\mu_0^2}{2} A_\mu A^\mu \right) = \frac{\mu_0^2}{2} \boxed{\frac{\partial (A_\mu A^\mu)}{\partial A_\nu}}$$

se puede reescribir como:

$$\begin{aligned} \frac{\partial}{\partial A_\nu} (A_\mu A^\mu) &= \eta^{\mu\alpha} \frac{\partial}{\partial A_\nu} (A_\mu A_\alpha) \\ &= \eta^{\mu\alpha} \left\{ \frac{\partial A_\mu}{\partial A_\nu} A_\alpha + A_\mu \frac{\partial A_\alpha}{\partial A_\nu} \right\} \quad / \quad \frac{\partial A_\mu}{\partial A_\nu} = \delta_\mu^\nu \\ &= \eta^{\mu\alpha} \left\{ \delta_\mu^\nu A_\alpha + A_\mu \delta_\alpha^\nu \right\} \\ &= \eta^{\mu\alpha} \delta_\mu^\nu A_\alpha + \eta^{\mu\alpha} A_\mu \delta_\alpha^\nu \\ &= \eta^{\nu\alpha} A_\alpha + \eta^{\mu\nu} A_\mu \\ &= A^\nu + A^\nu = 2A^\nu \end{aligned}$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial A_\nu} = \frac{\mu_0^2}{2} \cdot 2A^\nu = \frac{\mu_0^2}{1} A^\nu$$

para el segundo término:

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} = \frac{\partial \mathcal{L}}{\underbrace{\partial F_{\alpha\beta}}_1} \cdot \underbrace{\frac{\partial F_{\alpha\beta}}{\partial (\partial_\mu A_\nu)}}_2$$

$$\begin{aligned}
 ① \quad \frac{\partial}{\partial F_{\alpha\beta}} (F_{\rho\sigma} F^{\rho\sigma}) &= \frac{\partial F_{\rho\sigma}}{\partial F_{\alpha\beta}} \cdot F^{\rho\sigma} + F_{\rho\sigma} \frac{\partial F^{\rho\sigma}}{\partial F_{\alpha\beta}} \quad / \quad F^{\rho\sigma} = \eta^{\rho\lambda} \eta^{\sigma\kappa} F_{\lambda\kappa} \\
 &= \delta_{\rho}^{\alpha} \delta_{\sigma}^{\beta} \cdot F^{\rho\sigma} + F_{\rho\sigma} \eta^{\rho\alpha} \eta^{\sigma\beta} \\
 &= F^{\alpha\beta} + F^{\alpha\beta} \\
 &= 2F^{\alpha\beta} \quad / \quad -\frac{1}{4\mu_0} \\
 \frac{\partial \mathcal{L}}{\partial F_{\alpha\beta}} &= -\frac{1}{2\mu_0} F^{\alpha\beta}
 \end{aligned}$$

$$\begin{aligned}
 ② \quad \frac{\partial E_{\beta}}{\partial (\partial_{\mu} A_{\nu})} &= \frac{\partial (\partial_{\alpha} A_{\beta})}{\partial (\partial_{\mu} A_{\nu})} - \frac{\partial (\partial_{\beta} A_{\alpha})}{\partial (\partial_{\mu} A_{\nu})} \\
 &= \delta_{\alpha}^{\mu} \delta_{\beta}^{\nu} - \delta_{\beta}^{\mu} \delta_{\alpha}^{\nu}
 \end{aligned}$$

Uniendo lo anterior:

$$\begin{aligned}
 \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} A_{\nu})} &= -\frac{1}{2\mu_0} F^{\alpha\beta} (\delta_{\alpha}^{\mu} \delta_{\beta}^{\nu} - \delta_{\beta}^{\mu} \delta_{\alpha}^{\nu}) \\
 &= -\frac{1}{2\mu_0} (F^{\mu\nu} - F^{\nu\mu})
 \end{aligned}$$

$$\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} A_{\nu})} = -\frac{1}{2\mu_0} (F^{\mu\nu} + F^{\nu\mu}) = -\frac{1}{\mu_0} F^{\mu\nu}$$

uniendo todo en la ec. de Euler-Lagrange:

$$\frac{\mu_0^2}{\mu_0} A^{\nu} - J^{\nu} - \partial_{\mu} \left(-\frac{1}{\mu_0} F^{\mu\nu} \right) = 0$$

$$\begin{aligned}
 \frac{1}{\mu_0} \partial_{\mu} F^{\mu\nu} + \mu_0^2 A^{\nu} &= J^{\nu} \quad / \cdot \mu_0 \\
 \partial_{\mu} F^{\mu\nu} + \mu_0^2 A^{\nu} &= \mu_0 J^{\nu} \quad / \partial_{\nu}
 \end{aligned}$$

$$\partial_{\nu} \partial_{\mu} F^{\mu\nu} + \mu_0^2 \partial_{\nu} A^{\nu} = \mu_0 \partial_{\nu} J^{\nu}$$

$$\partial_{\nu} A^{\nu} = 0$$

también puede escribirse como: $\partial_{\mu} F^{\mu\nu} = \partial_{\mu} (\partial^{\mu} A^{\nu} - \partial^{\nu} A^{\mu})$

$$\begin{aligned}
 &= \square A^{\nu} - \partial^{\nu} (\partial_{\mu} A^{\mu}) \\
 \Rightarrow (\square + \mu_0^2) A^{\nu} &= \mu_0 J^{\nu}
 \end{aligned}$$

El término de masa rompe la invariancia de gauge porque

$$\begin{aligned}
 A_{\mu} A^{\mu} &\rightarrow (A_{\mu} + \partial_{\mu} \Lambda) (A^{\mu} + \partial^{\mu} \Lambda) \\
 &= A_{\mu} A^{\mu} + 2A^{\mu} \partial_{\mu} \Lambda + \partial_{\mu} \Lambda \partial^{\mu} \Lambda.
 \end{aligned}$$

Los términos adicionales no son, en general, una derivada total. Por lo tanto, la acción con masa no es invariante bajo $A_{\mu} \rightarrow A_{\mu} + \partial_{\mu} \Lambda$. Físicamente, la teoría de Maxwell describe un fotón sin masa con dos polarizaciones transversales, mientras que Proca describe un campo vectorial masivo con tres polarizaciones físicas.

2.4. Utilice el teorema de Noether y la invariancia bajo traslaciones espacio-temporales para construir el tensor de energía-momento canónico $T_{\text{can}}^{\mu\nu}$ del campo electromagnético y muestre que el objeto resultante no es simétrico ni invariante de gauge.

Considerar la lagrangiana de campo libre

Traslación infinitesimal: $x^\mu \rightarrow x'^\mu = x^\mu + a^\mu$, lo cual genera que $\delta A_\rho = -a^\nu \partial_\nu A_\rho$.

El tensor de energía-momento:

$$T^{\mu\nu} = \pi^{\mu\rho} \partial^\nu A_\rho - \eta^{\mu\nu} \mathcal{L}$$

Similar a lo ya calculado

donde el momento conjugado es: $\pi^{\mu\rho} = \frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\rho)} = -F^{\mu\rho}$

$$\Rightarrow T^{\mu\nu} = -F^{\mu\rho} \partial^\nu A_\rho + \frac{1}{4} \eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \quad (x) \quad / \quad \partial_\mu$$

$$\partial_\mu T^{\mu\nu} = - \underbrace{\partial_\mu F^{\mu\rho}}_0 \partial^\nu A_\rho - F^{\mu\rho} \partial_\mu \partial^\nu A_\rho + \frac{1}{4} \eta^{\mu\nu} \partial_\mu (F_{\alpha\beta} F^{\alpha\beta})$$

$$= -F^{\mu\rho} \partial_\mu \partial^\nu A_\rho + \frac{1}{4} \eta^{\mu\nu} (2F^{\alpha\beta} \partial^\nu F_{\alpha\beta})$$

$$= -F^{\mu\rho} \partial_\mu \partial^\nu A_\rho + \frac{1}{2} F^{\alpha\beta} (\partial^\nu \partial_\alpha A_\beta - \partial^\nu \partial_\beta A_\alpha) \rightarrow = F^{\alpha\beta} \partial^\nu \partial_\alpha A_\beta$$

$$= -F^{\mu\rho} \partial_\mu \partial^\nu A_\rho + F^{\alpha\beta} \partial^\nu \partial_\alpha A_\beta \quad / \quad (\alpha, \beta) \rightarrow (\mu, \rho)$$

$$= -F^{\mu\rho} \partial_\mu \partial^\nu A_\rho + F^{\mu\rho} \partial^\nu \partial_\mu A_\rho$$

$$\boxed{\partial_\mu T^{\mu\nu} = 0} \rightarrow \text{se demuestra la conservación.}$$

Una forma de mostrar la simetría de un tensor es que su componente antisimétrica sea nula, es decir

$$T^{\mu\nu} - T^{\nu\mu} = 0$$

usando lo obtenido en la ec. (x), vemos que:

$$T^{\mu\nu} - T^{\nu\mu} = -F^{\mu\rho} \partial^\nu A_\rho + F^{\nu\rho} \partial^\mu A_\rho$$

donde no necesariamente ambos términos se anulan, por lo que no podemos afirmar que el tensor de energía-momento necesariamente se anule.

Tampoco se cumple la invariancia de gauge